

STANDARD CLASS

Running Gemma 4 31B: locally and in the cloud

One 96 GB card locally, one H100 in the cloud. The model every FlatClaw tenant runs today. **Model:** Gemma 4 31B-IT · 31B dense · Apache 2.0 · 256K native · ~33 GB at FP8 + KV cache.

96 GB

VRAM, ONE CARD

FP8NATIVE ON BLACKWELL /
HOPPER**256K**

CONTEXT SERVED

8-12

CONCURRENT STREAMS

1

PROCESS PER CARD

LOCAL · BOUGHT ONCE

\$23,448The inference workstation as specced below;
Newegg street prices on October 7, 2026.

CLOUD · HELD WARM

\$2,000/moNorthflank nf-gpu-h100-80-1g, 1 × H100 80
GB; list price, 24/7, CPU and RAM bundled.

01 Local — The inference workstation

A tower that is a daily desktop, the internal FlatClaw node and a demo server, bought in three phases so nothing is rebought: the platform first, a used 48 GB card for real CUDA serving, the 96 GB card when the memory market allows.

PART	PICK · WHY IT'S HERE	STREET	BUY
GPU · MAIN PHASE 3	NVIDIA RTX PRO 6000 Blackwell, 96 GB (Workstation Edition) 96 GB of GDDR7 holds the 31B at FP8 with the full 256K window on one card, exactly how the hosted tenant runs it. Blower, display out, 600 W. The passive Server Edition is wrong for a tower.	\$16,700 in stock · list \$18,000	Newegg ↗
GPU · 2ND PHASE 2	NVIDIA RTX A6000, 48 GB (used) Phase-2 serving card: the 31B at 4-bit runs ~35–40 tok/s. End state, the utility lane: image, voice, embeddings and a small LLM resident with room to spare. Ampere has no FP8, so anything on it is quantized.	\$3,250 used market (eBay), \$3,000–3,500 in August 2026; Newegg does not list it	search ↗
GPU · BOOT PHASE 1	AMD Radeon RX 5700 XT, 8 GB (on hand) Free bootstrap and permanent display card: amdgpu drives the desktop so the NVIDIA cards stay headless with every byte of VRAM free.	\$0.00 already owned	—
CPU PHASE 1	AMD Ryzen 9 9900X, 12 cores Feeds both cards over PCIe 5.0 and runs the OS, SGLang and the desktop. Twelve cores is plenty; the 9950X only pays off for CPU-heavy ingestion.	\$379 in stock · list \$403	Newegg ↗
MAINBOARD PHASE 1	ASUS ProArt X870E-Creator WiFi Two PCIe 5.0 x16 slots (x8/x8), 10 GbE + 2.5 GbE, four M.2, USB4: workstation-grade AM5.	\$512 in stock	Newegg ↗
MEMORY PHASE 1	Corsair Vengeance 96 GB (2 × 48 GB) DDR5 Two DIMMs keep AM5 at full speed; 96 GB carries desktop, Docker, gateway and ingestion with headroom. Models live in VRAM. Newegg lists the 5600 kit today; take the 6000 CL30 kit when it is back.	\$1,425 out of stock	Newegg ↗
POWER PHASE 1	Corsair RM1200x Shift, 1200 W, 80+ Gold, ATX 3.1 Native 12V-2x6 for the Pro 6000, 8-pin PCIe for the second card; right-sized for two big cards at ~1.1 kW peak.	\$160 not listed on Newegg today; \$160 street	search ↗
CPU COOLER PHASE 1	Arctic Liquid Freezer III Pro 360 Holds the 120 W 9900X while the GPUs dump heat into the case; best-value 360 mm.	\$106 not listed on Newegg today; \$106 street	search ↗
CASE PHASE 1	Fractal Design North XL, Mesh, charcoal and walnut A tower that looks like furniture: 413 mm of GPU room, a 360 mm top radiator, mesh side feeding the blowers. Mesh, not glass.	\$166 in stock · list \$195	Newegg ↗
STORAGE PHASE 1	Samsung 9100 PRO 4 TB, NVMe Gen5 OS plus the model library (~1 TB system, ~3 TB weights); the board takes Gen5 natively.	\$750 in stock · list \$1,360	Newegg ↗
Phase 1 — Platform + the card on hand · Desktop, portal, a 7B model through Vulkan. Buy RAM and the SSD first; DDR5 is still rising.		\$3,498	
Phase 2 — + used A6000 · The 31B at 4-bit around 35–40 tok/s: demo-able, plus the whole utility lane on one card.		\$6,748	
Phase 3 — + RTX PRO 6000 · Production-identical FP8 at 256K; 144 GB of NVIDIA VRAM across the two cards.		\$23,448	
All-in, as specced · 10 parts, 6 priced live from newegg.com on October 7, 2026, the rest at reference prices		\$23,448	

Newegg street prices as read on October 7, 2026; the live table at flatclaw.org/builds refreshes hourly. Used-market and unlisted parts are marked. Verify at checkout.

02 Alternatives Newegg has today (not in the total)

PART	PICK · WHEN TO TAKE IT	STREET	BUY
GPU · ALT NOT IN TOTAL	NVIDIA RTX PRO 6000 Blackwell Max-Q, 96 GB Same 96 GB at 300 W with a quieter blower; a few percent slower. The right card when two must sit side by side (see the frontier build).	\$17,000 in stock	Newegg ↗
POWER · ALT NOT IN TOTAL	Seasonic PRIME TX-1600, 1600 W, 80+ Titanium In stock on Newegg today and enough for any two-card configuration; the same unit the frontier hybrid uses.	\$742 in stock	Newegg ↗
COOLER · ALT NOT IN TOTAL	Arctic Liquid Freezer III 420 A-RGB The 420 mm sibling Newegg has in stock; check the radiator fit for your case before swapping.	\$200 in stock	Newegg ↗

03 Cloud — the same checkpoint, one service

NORTHFLANK · RECOMMENDED · nf-gpu-h100-80-1g

\$2,000/mo · \$2.74/hr

1 × H100 80 GB · 80 GB. 26 vCPU and 234 GB RAM bundled with the GPU;
~\$2,000/month held warm 24/7 including the bundled compute. Cold boot
~10 minutes: image pull, weights fetched from the weights-server pod over
the internal network, SGLang load.

NORTHFLANK · nf-compute-400-16 (control)

\$144/mo

none. Portal plus one OpenClaw gateway per user: 4 vCPU / 16 GB carries
two or three users at 0.6–1.2 GB each with room to spare.

NORTHFLANK · nf-compute-50 + 200 GB volume

\$30/mo

none. weights-server: a CPU pod serving the tenancy's weights volume over
HTTP, because GPU plans cannot mount the volume directly.

ALL-IN

About \$2,000–2,200 a month per tenant, held warm. List prices;
committed-use terms on any cloud cut the GPU line.

THE SAME SHAPE ELSEWHERE

- ▶ **Azure** · NC H100 v5 — AKS node pool or a single GPU VM in the customer's subscription; same containers, same weights-volume pattern (Blob + blobfuse).
- ▶ **AWS** · p5 / g6e class — EKS or a single GPU instance in an account the customer owns; IAM + KMS, private VPC egress only.
- ▶ **Google Cloud** · A3 (H100) — GKE with an A3 node or a single GPU VM; Workload Identity, VPC Service Controls.

NOTES

- ▶ Flat per-tenant rate, not per-token metering: the GPU line dominates and everything else is under \$200.
- ▶ Capacity is peak concurrent sessions, not head count: one H100 sustains 8–12 concurrent streaming chats at FP8 with first tokens in 1–2 s.

04 Run it

```
# Card 0
CUDA_VISIBLE_DEVICES=0 python -m sglang.launch_server --model-path google/gemma-4-31b-it --quantization fp8 --tp 1 --context-length 262144 --tool-call-parser gemma4 --reasoning-parser gemma4 --port 30000 --mem-fraction-static 0.90
```

```
# Card 1
ComfyUI :8188 · voice :8001 · text-embeddings-router :8080 · optional 7–8B at 4-bit on its own port
```

- ▶ Both cards flat out: Pro 6000 600 W + A6000 300 W + platform ~200 W ≈ 1.1 kW, about 9 A at 120 V. Fine on a 15 A circuit; a dedicated 20 A circuit leaves room for a second A6000.
- ▶ Ubuntu 24.04 Desktop: amdgpu drives the display card and Wayland; NVIDIA R570+ / CUDA 12.8+ (sm_120) runs compute headless. Serve from containers and hold driver updates before demo days.
- ▶ One process per card, each pinned with CUDA_VISIBLE_DEVICES, behind one OpenAI-compatible front door. The flagship never reloads; the utility card hosts image, voice, embeddings and a small LLM.
- ▶ Run the OpenClaw gateway and the Portal on the same box for a fully local demo, or point a separate control machine at its endpoint; both work.